Collections Analytics — Data Architecture

From raw data to actionable insights

This document walks through how data moves through the collections analytics pipeline, from the moment it lands in its raw form to the point it shows up as a decision-ready dashboard for leadership. The pipeline is built as seven layers, each one cleaning up or adding value to the data before passing it downstream, with data quality and governance checks running alongside every step.

1. Raw

Everything starts here. Accounts, calls, payments, campaigns, digital activity, field data, PTPs, and other source files land in this layer exactly as they come in, with no modification. The idea is to keep an immutable, as-is copy of the source data, so there's always a clean starting point to fall back on if anything downstream needs to be reprocessed.

2. Staging & Validation

Before anything gets used, it passes through a set of basic checks — schema validation, data type checks, mandatory field validation, and soft referential integrity checks. Raw data is preserved alongside these checks rather than overwritten, and this is where data contracts are enforced for the first time.

3. Clean

This layer is where the data actually gets fixed. Duplicate payments and calls are removed, IDs get resolved across agents and accounts, timestamps are normalized to a consistent timezone (Asia/Kolkata), missing values are handled, and historical corrections are applied. By the end of this stage, the data is conformed and consistent enough to be trusted.

4. Golden (Canonical)

This is the single source of truth for the business. Canonical versions of accounts, customers, agents, and deduplicated payments live here, along with account-month facts. Anything built after this layer pulls from here rather than going back to raw or clean data.

5. Feature

Here, the canonical data is shaped into business-ready entities — risk and DPD features, campaign features, channel features, agent features, contact and PTP features, targeting features, and portfolio attributes. This is effectively the feature store that everything analytical gets built on top of.

6. Metrics & Aggregations

This layer turns features into the actual numbers people care about — raw and deduplicated recovery, recovery per targeted account, contact rate, recovery per call, PTP creation rate, channel conversion, strategy performance, and ROI and other financial metrics. These are the curated metrics that feed directly into analytics and reporting.

7. Power BI Dashboard

The final layer is where it all comes together for decision-makers — an executive dashboard covering trends and drivers, strategy and channel performance, data quality views, and the investment decision around the Rs. 10 Cr allocation. This is the layer built for leadership to act on, not to dig through raw numbers.

Data Quality & Governance

Running alongside every layer, rather than sitting at the end of the pipeline, is a governance layer that keeps the whole thing trustworthy:

- **Data Contracts** — Schema, data types, mandatory fields, and acceptance rules.
- **Keys & Integrity** — PK/FK checks, referential integrity, and uniqueness checks.
- **Data Quality Checks** — Duplicate checks, missing data, range and format validations.
- **Lineage & Audit** — End-to-end data lineage, audit trails, and change tracking.
- **Incremental Processing** — Daily incremental loads, watermarking, and late-arriving data handling.
- **Backfills & Corrections** — Historical backfills, reprocessing framework, and versioning.
- **Monitoring & Alerts** — Pipeline monitoring, DQ alerts, anomaly detection, and SLA monitoring.

Feedback Loop

Insights generated at the dashboard layer don't just sit there — they loop back into the system. Findings from the dashboard inform better targeting, strategy refinement, and ongoing data quality improvements, which in turn feed back into the governance layer and the pipeline as a whole.

Legend Reference

Stage	Category
1. Raw	Ingestion
2. Staging	Validation
3. Clean	Cleansing
4. Golden	Trusted Data
5. Feature	Feature Engineering
6. Metrics	Analytics
7. Power BI Dashboard	Visualization